

Choose the correct answer and fill it in the table below

(30 Pts)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
d	d	b	b	d	a	d	b	b	b
Q11	Q12	Q13	Q14	Q15					
d	b	b	c	c					

1- If the general solution of a D.E. is

$$y = c_1 x^{-1} + x^{-2}(c_2 \cos(3 \ln x) + c_3 \sin(3 \ln x)), \text{ then the D.E. is}$$

a) $x^3 y''' - 8x^2 y'' - 23xy' + 13y = 0$

b) $x^3 y''' - 8x^2 y'' + 23xy' + 13y = 0$

c) $x^3 y''' + 8x^2 y'' - 23xy' + 13y = 0$

✓ d) $x^3 y''' + 8x^2 y'' + 23xy' + 13y = 0$

2- To solve the nonlinear D.E. $y' - 2x^2 y = 3x^4 y^{-2}, y > 0$, we convert it to

L.D.E., then the integrating factor for the resulting Linear D.E. is $\mu =$

a) $e^{\frac{2}{3}x^3}$

b) e^{-x^3}

c) $e^{\frac{-2}{3}x^3}$

✓ d) e^{-2x^3}

3- Suppose $u = \frac{y}{x}$ then the general solution of $y' = \frac{y}{x} + e^{\frac{y}{x}}$ is $\ln x =$

a) $-e^u + c$

✓ b) $-e^{-u} + c$

c) $e^{-u} + c$

d) $e^u + c$

4- If $y_1 = x+1$ is a solution of $(x^2 + 2x+1)y'' + (x+1)y' + Ay = 0$, then $y_2 =$

a) $\frac{-1}{4(x+1)^3}$

✓ b) $\frac{-1}{2(x+1)}$

c) $(x+1)\ln(x+1)$

d) $x^2 + x$

5- The largest possible interval at which the I.V.P.

$$\tan(x)y'' + x^2 y' + 4y = \frac{1}{x^2 - 4}, y(3) = 2, y'(3) = 1 \text{ has a unique solution is}$$

a) $\left(\frac{\pi}{2}, \pi\right)$

b) $\left(2, \frac{3\pi}{2}\right)$

c) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

✓ d) $(2, \pi)$

6- The general solution of $y^{(4)} - 2y''' = 0$ is $y =$

✓ a) $c_1 + c_2 x + c_3 x^2 + c_4 e^{2x}$

b) $c_1 + c_2 x + c_3 e^{2x}$

c) $c_1 + c_2 x + c_3 e^{-2x}$

d) none

7- Suppose that the characteristic equation of a homogeneous L.D.E. with constant coefficients is $\lambda^5(\lambda-2)^4(\lambda^2+1)^3=0$, then ONE of the following is NOT a solution of the D.E.

- a) $x^4 - 3x^2 e^{2x}$ b) $x - x^3 \cos x$ c) $2x^3 e^{2x} + x \sin x$ d) 9

8- Which of the following is a solution to $xy' - y = 2x^3$

- a) x^2 ☒ b) x^3 c) e^x d) x

9- If $y' + p(x)y = q(x)y$, then the general solution will be of the form $y =$

- ☒ a) $e^{\int (q(x)-p(x))dx} + c$ ☒ b) $e^{\int (p(x)-q(x))dx} + c$ c) $\int (q(x)-p(x))dx + c$ d) none

10- One of the following is a second order nonlinear D.E.

- a) $\ln(y)y' + y = x$ ☒ b) $xy'' + y' = ye^x$ c) $\sin(x)y'' + yy' = 1$ d) none

11- The value of m that makes $y = x^m$ a solution to $x^2 y' - 3xy = 0$ is

- a) $-\frac{3}{2}$ b) 2 c) -3 ☒ d) 3

12- The particular solution of $(2xy - \sec^2 x)dx + (x^2 + 2y)dy = 0, y(0) = 2$, is $y =$

- a) $x^2 y - \tan x - y^2 = 4$ ☒ b) $x^2 y - \tan x + y^2 = 4$ c) $x^2 y + \tan x + y^2 = 4$ d) none

13- If the integrating factor of $f(y)dx + (x + y^2)dy = 0$ is $\mu = y$, then $f(y) =$

- a) $\frac{y}{2} + \frac{c}{y}$ ☒ b) $\frac{y}{2} + \frac{g(x)}{y}$ c) $\frac{y^2}{2} + c$ d) $\frac{y^2}{2} + g(x)$

14- The general solution of the exact D.E $(2x + y \cos(xy))dx + x \cos(xy)dy = 0$ is

- a) $\sin(xy) + 2x^2 = c$ b) $\sin(xy) - x^2 = c$ ☒ c) $\sin(xy) + x^2 = c$ d) none

15- The general solution of $(1 + 3y^3)y' = y \cos x$ is

- a) $\ln y + y^3 = \sin x + c$ b) $\ln y + y^3 = -\sin x + c$ ☒ c) $\ln y + y^2 = \sin x + c$ d) none

$$\int \frac{1}{y} + 3y^2 dy = \int \cos x dx$$

GOOD LUCK...

5- The suitable form for y_p of $y'' + y' - 4y = 2e^{-x} + \cos x$ is

- a) $Ae^{-x} + B \cos x + C \sin x$ b) $Axe^{-x} + B \cos x + C \sin x$
 c) $Axe^{-x} + B \cos x$ d) $x e^{-x} (A + Bx) + C \sin x + D \cos x$

6- The Laplace transform of $\frac{1}{t^2} \delta(t-3)$ equals

- a) $9e^{-3s}$ b) $9e^{-3s}$ c) $-3e^{-3s}$ d) $3e^{-3s}$

7- If the variation of parameters method is used to find $y_p = y_1 u_1 + y_2 u_2$ for $x^2 y'' - 3xy' + 3y = 2x^4 e^x$, such that $y_1 = x$, $y_2 = x^2$ then $u_2 =$

- a) $-x^3 e^x + 2x e^x - 2e^x$ b) $-e^{-x}$ c) $-2e^x$ d) e^x

8- The singular solution of $y = xp + (4-p)^2$, where $p = y'$, is $y =$

- a) $-4x + \frac{3x^2}{4}$ b) $4x - \frac{x^2}{4}$ c) $-4x + \frac{x^2}{4}$ d) none

9- The inverse Laplace of $\left\{ \frac{s^2 - 1}{(s+1)^2} \right\}$ equals

- a) $\frac{1}{2} t^2 e^{-t} - \frac{1}{3} t^3 e^{-t}$ b) $t^2 e^{-t} - \frac{1}{2} t^3 e^{-t}$ c) $\frac{1}{2} t^3 e^{-t} - \frac{1}{6} t^4 e^{-t}$ d) none

10- Assume that $F(s) = t \left\{ \sinh(\ln t) + \frac{1}{2t} \right\}$, then $F(0.5) =$

- a) -0.5 b) 4 c) -2 d) 2

11- The orthogonal trajectories of the family of curves $e^x + e^{-x} = c$ are

- a) $e^x + e^{-x} = c$ b) $e^x + e^x = c$ c) $e^x - e^{-x} = c$ d) $e^{-x} + e^x = c$

12- The general solution of $y'' - 6y' + 5y = 0$ is

- a) $c_1 e^x + c_2 e^{3x} + c_3 e^{4x}$ b) $c_1 e^{-x} + c_2 e^{3x} + c_3 e^{4x}$
 c) $c_1 e^x + c_2 e^{-3x} + c_3 e^{-4x}$ d) $c_1 e^{-x} + c_2 e^{3x} + c_3 e^{-4x}$



Fill in the table below with the correct answer

(50 Pts)

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Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
Q23	Q24	Q25								

1- If $y'' = \delta(t+1)$, $y(0) = 0$, $y'(0) = -1$, such that $\ell\{y(t)\} = Y(s)$, then $Y(1) =$

- a) $e^{-1} - 1$ b) $e - 1$ c) $e + 1$ d) $e^{-1} + 1$

2- The particular solution of $yy' = x+1$, $y(0) = 1$ is $y^2 =$

- a) $x^2 + 2x - 1$ b) $x^2 + 2x + 2$ c) $x^2 + 2x$ d) $x^2 + 2x + 1$

3- If $(y^3 + kxy^4 - 2x)dx + (3xy^3 + 20x^2y^3)dy = 0$, then the value of k that makes the D.E exact is

- a) -10 b) 4 c) -40 d) 20

4- If $X_1 = \begin{pmatrix} e^{-t} \\ e^t \end{pmatrix}$, $X_2 = \begin{pmatrix} 3e^{-5t} \\ -e^{-5t} \end{pmatrix}$ are two solutions of the homogeneous part of

$X' = AX + \begin{pmatrix} 6 \\ 1 \end{pmatrix}$ such that the particular solution $X_p = X_1u_1 + X_2u_2$, then $u_1 =$

- a) $\frac{9}{4}e^t$ b) $\frac{9}{4}e^{-t}$ c) $\frac{1}{4}e^{5t}$ d) $\frac{25}{4}e^{5t}$

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DEC 14, 2016

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13- If the G.S. of $x^3 y'' + ax^2 y' + bxy + cy = 0$ is $y = c_1 x + c_2 x \ln x + c_3 x^2$ then $a =$

- a) 1 b) 0 c) -1 d) 2.

14- The general solution of $(x \tan(\frac{y}{x}) + y)dx - xdy = 0$ is

- a) $x \cos(\frac{y}{x}) = c$ b) $x \sin(\frac{y}{x}) = c$ c) $x \csc(\frac{y}{x}) = c$ d) $x \sec(\frac{y}{x}) = c$.

15- By using $y = \sum_{n=0}^{\infty} a_n x^n$ as the general solution of $y'' + 4y = 0$, then $a_2 =$

- a) $-2a_0$ b) $-\frac{2}{3}a_0$ c) $-4a_0$ d) $-\frac{1}{3}a_0$.

16- If $Y(s) = \mathcal{L}\{y(t)\} = \frac{s}{s+1}$, and $\mathcal{L}\{y'(t)\} = Y_1(s)$, then $Y_1(1) =$

- a) $-\frac{3}{4}$ b) $-\frac{1}{4}$ c) $-\frac{1}{2}$ d) $-\frac{5}{4}$

17- If $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{2s}}{s^2} - \frac{1}{s^2}\right\}$, then $f(2) =$

- a) -2 b) 2.5 c) 2 d) 0

18- The integrating factor of $xy' + 2xy + 2y = e^{-x} \sin(2x)$ is

- a) $2 \ln(x) + x^2$ b) xe^x c) xe^{2x} d) $x^2 e^{2x}$

19- If $F(s) = \mathcal{L}\{e^{-2t} f(t)\} = e^{-2(s+3)}$, then $f(t) =$

- a) $u(t+2)$ b) $\delta(t-2)$ c) $\delta(t+2)$ d) $u(t-2)$